

SUPPLY CHAIN DELIVERY PERFORMANCE REPORT

End to End Order Fulfillment Analysis & Predictive Insights

Core business finding

54.71% of the 172,765 analyzed orders were delivered late. The report identifies shipping-mode performance as the most differentiated operational issue and evaluates the financial and operational implications.

Prepared by Ayush
Based on the submitted supply chain analysis

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1. Executive Summary

This report presents a comprehensive analysis of delivery operations for a global e-commerce company managing end-to-end order fulfillment across multiple regions. The analysis covers 172,765 orders from January 2015 through January 2018 and focuses on late-delivery drivers, financial impact and improvement opportunities.

The headline finding is that 54.71% of orders were delivered late, while the on-time delivery rate was 45.29%. Delayed orders were associated with \$2.1M in profit at risk, compared with \$7.5M in total profit from profitable orders. The Random Forest model achieved 74% accuracy in identifying orders at risk of late delivery.

Key metric	Reported result
Orders analyzed	172,765
Late deliveries	94,523
Late delivery rate	54.71%
On-time delivery rate	45.29%
Total profit from profitable orders	\$7.5M
Profit at risk on delayed orders	\$2.1M
90th-percentile delay	3 days
Random Forest accuracy	74%

Interview-ready takeaway

The analysis connects operational performance with business impact: delivery delays are not only a customer-experience problem, but also a profitability and service-reliability problem. The report then moves from diagnosis to prioritized recommendations.

2. Business Context and Objectives

The company operates a global e-commerce platform selling sporting goods, fitness equipment, outdoor gear, footwear and apparel across multiple international regions. The core business problem is that actual shipping times frequently deviate from scheduled delivery windows, reducing customer trust, weakening order profitability and limiting the reliability of delivery commitments.

Analytical objectives

Objective	Purpose
Assess delivery performance	Understand performance by region, shipping mode, time, customer segment and other operational dimensions.
Quantify financial impact	Measure how delivery delays relate to order-level profitability.
Identify bottlenecks	Locate the operational categories with the highest delay rates.
Build a predictive model	Flag high-risk orders before they become late.
Recommend action	Translate findings into prioritized steps to improve delivery and profitability.

3. Performance Dashboard

The following indicators establish the performance baseline from the cleaned dataset.

Indicator	Value	Interpretation
Total orders	172,765	Scope of the analysis
Late deliveries	94,523	More than half of all analyzed orders
Late delivery rate	54.71%	Primary performance concern
On-time delivery rate	45.29%	Less than half of orders met the delivery promise
Total profit (profitable orders)	\$7.5M	Profit base exposed to service performance
Profit at risk (delayed orders)	\$2.1M	Financial impact associated with delayed orders
90th-percentile delay	3 days	Delays are concentrated within a short range
Average order profit	\$22.03	Reported average order-level profit

4. Profitability Analysis

4.1 Profitability distribution

Order-level profitability was classified into three tiers using Order Profit Per Order. The source analysis reports that 80.7% of orders were profitable, 18.7% were loss-making and 0.6% were at breakeven. The loss-making share represents a meaningful drag and is reported as disproportionately concentrated among delayed shipments.

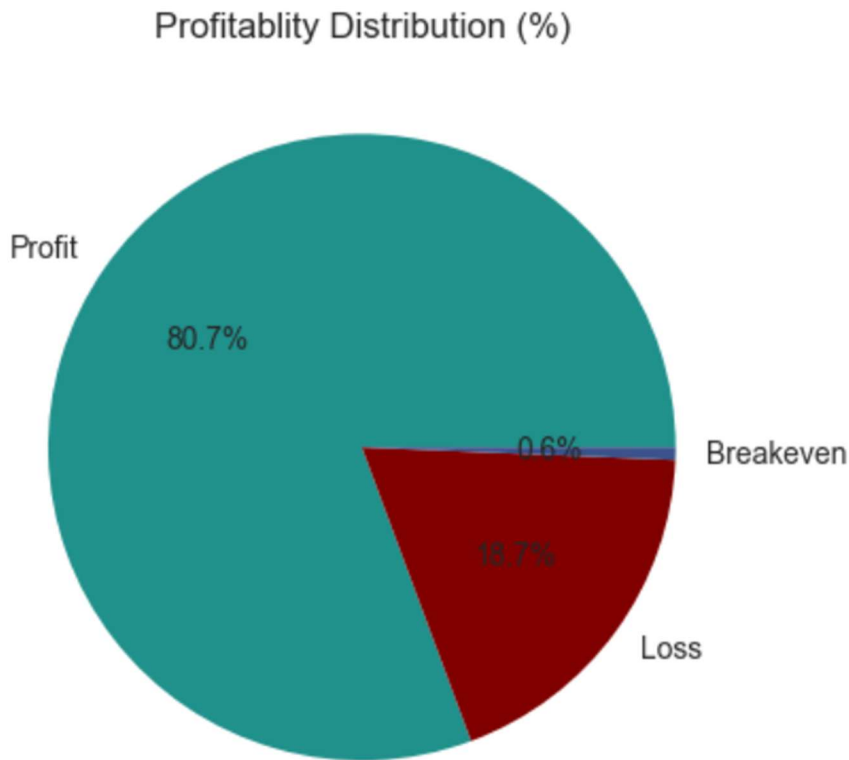


Figure 1. Profitability distribution from the submitted analysis.

4.2 Delay distribution and profit by delay days

The largest delay cohort is one day late, representing 31.0% of all orders. Orders delayed by one to four days account for 54.7% of total order volume, aligning with the overall late-delivery rate. The source analysis also reports that mean profit per order remains approximately \$20–\$23 across delay levels; therefore, the reported profitability exposure is driven primarily by the volume of delayed orders rather than a sharp change in individual order economics.



Figure 2. Delay distribution and profit analysis by delay days from the submitted analysis.

5. Bottleneck Detection

Delay percentages were compared across six operational dimensions: region, customer segment, shipping mode, order status, payment type and department. Shipping mode shows the strongest differentiation and is therefore identified in the source report as the primary operational lever.

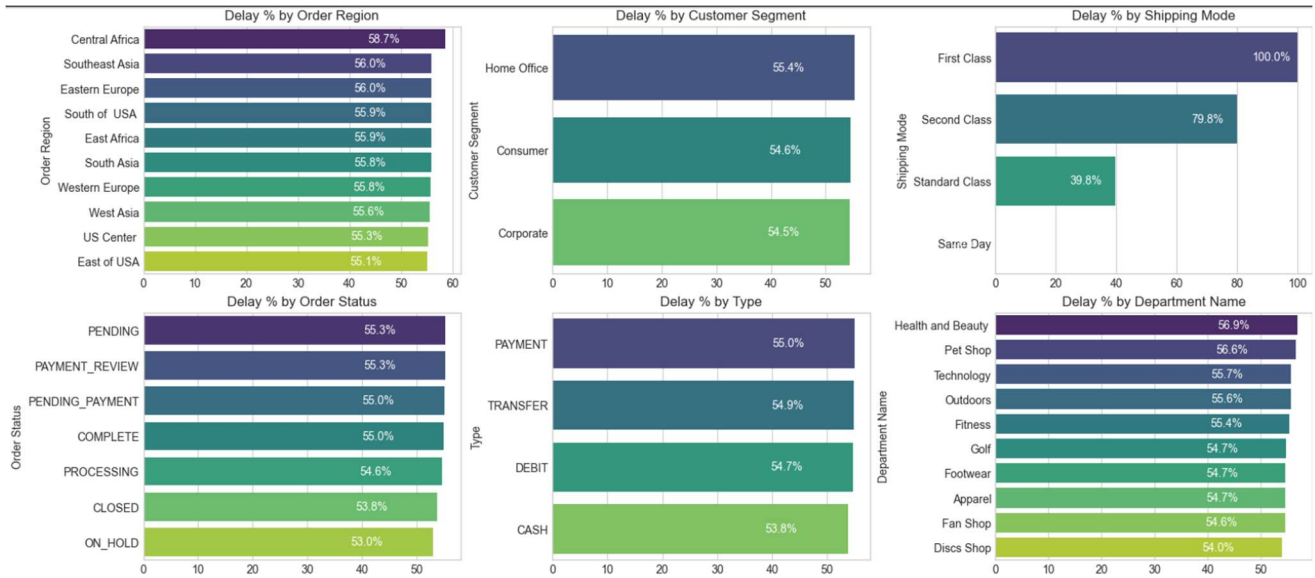


Figure 3. Delay percentage across the six operational dimensions.

Dimension	Reported finding	Management implication
Shipping mode	First Class 100.0%; Second Class 79.8%; Standard Class 39.8%; Same Day 0.0%	Validate mode assignment logic and carrier/service performance first.
Region	Central Africa highest at 58.7%; regions cluster around 55–59%	Indicates a broad operational issue rather than only a localized failure.
Customer segment	Consumer, Corporate and Home Office approximately 54.5–55.4%	Segment differences are limited.
Department	Health & Beauty 56.9%; Pet Shop 56.6%	Conduct an inventory and carrier audit for these categories.
Payment/order status	PAYMENT_REVIEW 55.3% overall and 80.0% in Central Africa; PENDING_PAYMENT 55.0%	Payment-processing friction is a secondary bottleneck.

6. Root Cause Analysis

The source report selected Central Africa for a deep-dive analysis because it was identified as the highest-delay region at 58.7%. The top reported drivers within the regional analysis include First Class shipping, PAYMENT_REVIEW status, Second Class shipping, CASH payment type, CLOSED and ON_HOLD statuses, Fitness, PENDING status, TRANSFER payment type and Apparel.

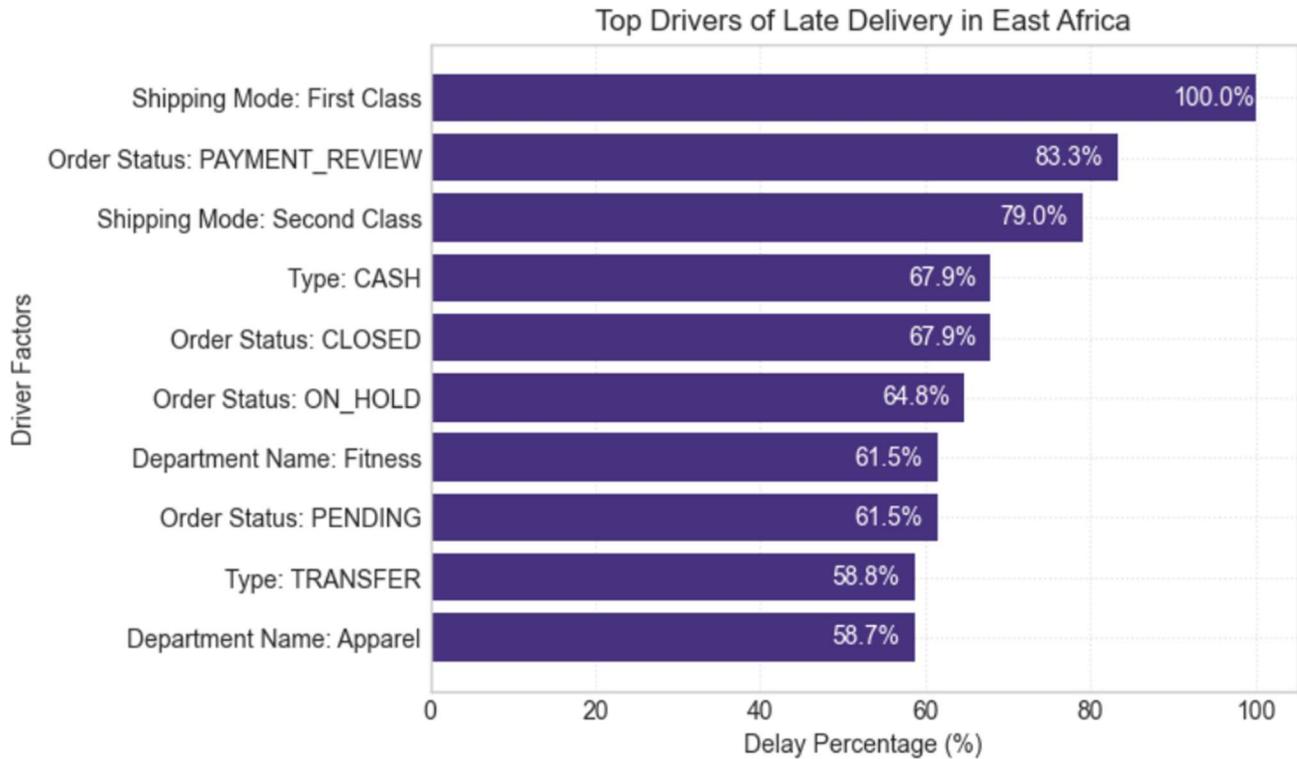


Figure 4. Top drivers of late delivery in the regional deep-dive from the submitted analysis.

The findings indicate that the most important root-cause themes are shipping-mode performance, payment or order-status friction and category-level execution. The source report recommends a warehouse-level review of stock availability, pick-and-pack times and carrier assignment for high-delay categories.

7. Time-Based Delay Patterns

The analysis examined month of year, day of week and hour of day. Overall delay rates were relatively stable across these dimensions, ranging from approximately 53% to 57%, while selected peaks identify opportunities for capacity planning and cutoff review.

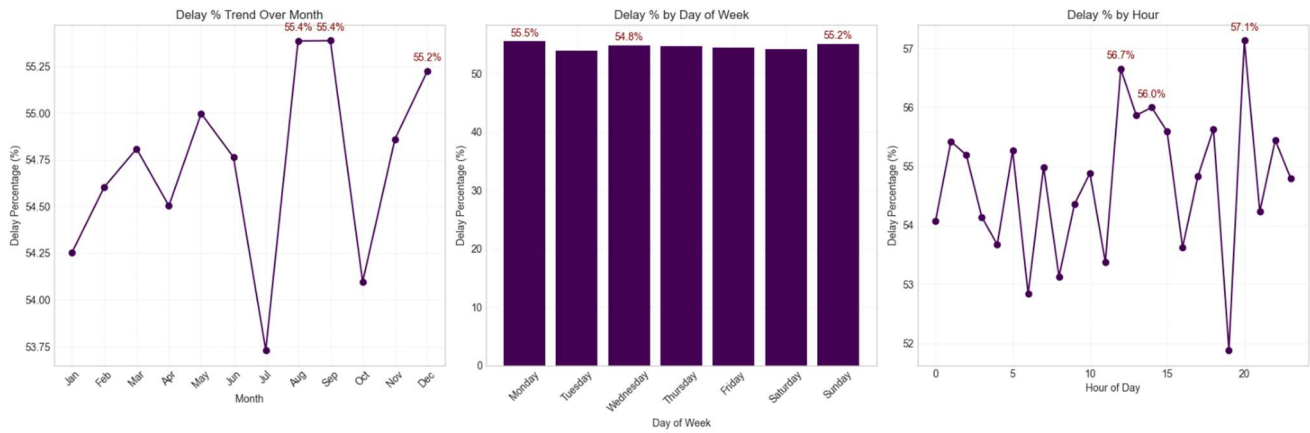


Figure 5. Delay percentage by month, day of week and hour from the submitted analysis.

Time dimension	Reported pattern	Interpretation in the source report
Month	August and September 55.4%; December 55.2%; July approximately 53.75%	Seasonal variation may reflect promotions and holiday volume.
Day of week	Monday 55.5% and Sunday 55.2%; spread below one percentage point	Day of week is not a meaningful primary intervention lever.
Hour of day	Reported peaks include hour 21 at 57.1%, hour 13 at 56.7% and hour 12 at 56.0%	Review processing cutoffs and peak-hour capacity.

8. Machine Learning Model Results

A supervised classification model was built to predict whether an order would be delivered late, using `Late_delivery_risk = 1` as the target. The pipeline included frequency encoding of categorical variables, a stratified train/test split, SMOTE oversampling to address class imbalance and Random Forest classification.

SMOTE stage	Class 0: On-time	Class 1: Late
Before SMOTE (training set)	59,030	79,182
After SMOTE (training set)	79,182	79,182

Random Forest test-set performance: 34,553 records.

Metric	On-time class	Late class	Overall / weighted
Precision	0.68	0.78	0.74
Recall	0.72	0.75	0.74
F1-score	0.70	0.77	0.74
Accuracy	—	—	0.74
Support	14,758	19,795	34,553

Model interpretation

The model correctly classifies approximately three out of four test-set orders. Precision of 0.78 for late orders means that a predicted late order was correct 78% of the time in the reported test results. Recall of 0.75 indicates that the model captured 75% of actual late orders.

9. Strategic Recommendations

The recommendations below are reproduced and professionally structured from the submitted analysis. They are grouped by the priority levels defined in the source report.

Priority	Action	Source rationale
CRITICAL	Audit First Class and Second Class shipping capacity	First Class has a 100% delay rate and Second Class 79.8%; conduct an immediate carrier SLA audit and review whether suspension or repricing is required.
HIGH	Deploy the predictive alert system	Use the Random Forest model to flag high-risk orders at confirmation and trigger proactive communication and priority handling.
HIGH	Resolve payment-processing bottlenecks	Introduce automated escalation for stalled payment reviews exceeding two hours.
MEDIUM	Develop seasonal surge-capacity plans	Plan for August, October and December peaks through temporary logistics partnerships, inventory pre-positioning and adjusted delivery windows.
MEDIUM	Default to Standard Class for eligible orders	Re-evaluate mode assignment logic because Standard Class has a reported 39.8% delay rate.
MEDIUM	Investigate high-delay departments in Africa	Audit Outdoors and Golf in Central Africa, including stock availability, pick-and-pack time and carrier assignment.
LOW	Reduce the loss-making order share	Review pricing and discounts together with delay patterns to identify compounded margin problems.
LOW	Continuously retrain the predictive model	Retrain quarterly; explore carrier, weather and warehouse-utilization features and target recall above 80%.

10. Conclusion

The analysis identifies a clear and urgent performance gap: 54.71% of orders failed to meet their promised delivery windows, with \$2.1M in profit at risk on delayed orders. Shipping-mode performance is the dominant operational signal, while payment-processing friction, seasonal peaks and category-level differences provide additional areas for action.

The Random Forest model provides a practical predictive capability, achieving 74% accuracy and 0.78 precision on late-order predictions in the reported test results. Combined with the operational findings, the report provides a data-driven basis for improving delivery reliability, protecting the \$7.5M profitable-order base and reducing loss-making orders.

Priority area	Current state	Target state
Late delivery rate	54.71%	Below 30% within 12 months
First Class on-time rate	0%	Above 80%
Second Class on-time rate	20.2%	Above 60%
Predictive model accuracy	74%	Above 82% with enhanced model
Loss-making orders	18.7%	Below 12%
Profit at risk	\$2.1M	Reduce by 40%

Report scope note

This professional revision improves structure, language and presentation using only the content and visuals supplied in the original report. No new external facts or independent analysis have been added.

End of report